

Team Breakout Handout

Deadline to submit: 12th Sept 11:59pm PT

Meet your team, pick one use case, and sketch an end to end AI agent design.

Submission form: <https://forms.gle/E6cKoFUJxV5g86EC7>

1 Who is on this team

Add everyone's full name and email. Leave the last rows for members who join your team later.

#	FULL NAME	EMAIL
1	Gnana Sudheer Gavarraju	sudheeronsocial@gmail.com
2	—	—
3	—	—
4	—	—
5	—	—
6	—	—
7	—	—

Point person (owns this doc and coordinates the team): **Gnana Sudheer Gavarraju**

2 Warm up: go around the room

Icebreaker. Tell us your name, where you are joining from, and the one boring or repetitive task you would happily hand off to an AI agent tomorrow if you could.

Sudheer, joining from **Vijayawada** — would happily hand off: “Summarizing my emails and tracking/planning my monthly spend” to an AI agent.

3 Pick a topic and sketch the design

Talk through these three questions together, then jot your thinking below.

Q1. Pick the use case.

Health Sentinel — a guardrailed multi-agent nutrition & lifestyle assistant. Turning meals, wearable data, medical documents, and everyday spending/SMS signals into safe, explainable health guidance is slow and error-prone today: people either get a generic chatbot that will happily invent a confident-sounding medical claim, or no synthesis at all across scattered sources (a fitness app, a lab report PDF, a bank statement, a calendar). The team found this compelling specifically because the failure mode (a confidently wrong “you’re fine” or a fabricated diagnosis) is genuinely high-stakes, which makes it a strong test of guardrails and evals — not just a demo of agent orchestration.

Q2. Knowledge and tools.

Knows (RAG): the user’s own medical documents (blood reports, medical history) are chunked, embedded, and retrieved per-user with a fused dense + lexical score; the agent refuses rather than answers when retrieval evidence is weak, instead of fabricating a medical fact.

Does (tools/actions): simulated MCP connectors stand in for real integrations — an iWatch-style connector for sleep/activity, an SMS connector that parses order/gym/health messages (user-confirmed via checkboxes before use), a

calendar connector for a stress signal, and a lab-report upload path that extracts glucose/weight readings into a small SQLite time series for trend reasoning.

Q3. Autonomy and evals.

Autonomy vs. workflow: mostly a fixed workflow, deliberately. The graph topology (consent → rate limit → meal/context agents → prediction → guardrail gate → optional human review → recommendation → deterministic verifier → advisory critic → finalize) is hard-coded; the LLM is free to reason only inside narrow, schema-validated steps (what to predict, how to phrase a recommendation), never over whether a safety gate applies. A deterministic, pure-Python verifier is the authoritative pass/fail check; an LLM critic is advisory-only and can never override it.

Evals: a two-tier eval suite — 40+ deterministic offline checks (guardrail thresholds, allergen matching, trend classification, category allowlist) that need no LLM call and gate CI on every push, plus a full end-to-end suite (RAG refusal/answer, human-in-the-loop triggering, cross-user isolation) and an adversarial red-team suite (direct diagnosis requests, prompt injection, allergen smuggling, confidence inflation) with a 100% block-rate bar on the hard cases. 59/59 checks pass today.

4 Our project

Use case, where RAG fits, tools the agent calls, autonomy vs workflow, and how you would evaluate it.

- **Use case:** Health Sentinel — guardrailed nutrition & lifestyle coaching that never diagnoses, built as a LangGraph StateGraph rather than a free-form agent loop.
- **Where RAG fits:** the user's own medical documents only, retrieved with a per-user metadata filter and a hard refusal threshold — grounding, not general medical knowledge.
- **Tools the agent calls:** simulated wearable (iWatch), SMS, calendar, and lab-report-upload connectors; every SMS-derived signal is confirmed by the user via checkboxes before it's used.
- **Autonomy vs. workflow:** fixed graph topology with deterministic gates; the LLM proposes, code (allowlists, bounds checks, the authoritative verifier) disposes.
- **How we'd evaluate it:** deterministic unit evals for every guardrail + an adversarial red-team suite with a hard 100% block-rate bar for diagnosis/prompt-injection attempts, gating CI before any change ships.

Next step. How and when we'll meet over the next 3 weeks: Not yet scheduled (TBD).

Please submit your projects here: <https://forms.gle/E6cKoFUJxV5g86EC7>

In scope: RAG, tool calling, autonomy vs workflow, evals. · **Not required:** fine tuning, AI security. · **Build over 3 weeks** (optional). · **Demo day** July 12 to 13.